Hartshorne Chapter 2 Section 1 Practice

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1 D

Problem 1

> Let A be an abelian group, and define the *constant presheaf* associated to A on the topological space X to be the presheaf $U \mapsto A$ for all $U \neq \emptyset$, with the restriction maps be the identity. Show that the constant sheaf is the sheafification this presheaf.

Solution: Here, can assume $A \neq 0$ (as the zero case is always a sheaf regardless, in fact the initial / final object in $\mathsf{Sh}(X)$).

Let $C_A : O(X)^{\text{op}} \to \mathsf{Ab}$ be the constant presheaf $C_A(U) = A$ for all nonempty open subset $U \subseteq X$ (and $C_A(\emptyset) = 0$). Even though the condition is stronger than the one given in Krishna's class, it's still not a sheaf in general: Given $U, V \subseteq X$ two nonempty open subsets such that $U \cap V = \emptyset$. Then, for any distinct $a, b \in A$, let $a \in F(U)$ and $b \in F(V)$, then their restriction onto the intersection is $a|_{U\cap V} = 0 = b|_{U\cap V}$, which they always agree on the intersection.

Yet, there doesn't exist any $c \in F(U \cup V)$, such that $c|_U = a$ and $c|_V = b$ (as the restriction should keep c). So, in general the gluing condition fails.

Now, to consider its sheafification, let's first talk about its stalks: Given any $x \in X$, notice the stalk $(C_A)_x = A$,since given any open neighborhood $U, V \subseteq X$ of x, any $a \in F(U) = A$ and $b \in F(V) = A$ are equivalent iff there exists $c \in F(W)$ for some open subset $W \subseteq U \cap V$, such that $a|_W = c = b|_W$. Yet, with all restriction of nonempty open subsets being identity, one has $a = b = c$. Now, notice that any open neighborhood of x has nontrivial intersection, so one has $a \in F(U)$ and $b \in F(V)$ satisfies $a \sim b$ iff $a = b$, hence the direct limit is A itself.

So, the sheafification $(C_A)^{\mathsf{Sh}}(U) = \left\{\text{functions } s : U \to \bigsqcup_{x\in U} F_x = \bigsqcup_{x\in U} A\right\}$ can be thought of as all functions $s : U \to A$, such that all $x \in U$ has some open neighborhood $x \in V \subseteq U$ and element $t \in F(V) = A$, such that all $y \in V$ satisfies $s(y) = t_y = t$.

Notice that this implies all functions in $(C_A)^{\mathsf{Sh}}(U)$ is locally constant. Which, we claim that if A endows with discrete topology, then the set $C_A^{\mathsf{Sh}}(U)$ is precisely $\mathrm{Hom}_{\mathsf{Top}}(U, A)$:

- Suppose $s \in C_A^{\mathsf{Sh}}(U)$, then for any element $a \subseteq A$, one has any $x \in s^{-1}(a)$ having some open neighborhood $x \in V_x \subseteq X$, and some $t \in F(V_x)$, such that $s(y) = t_y = t \in A$ for all $y \in V$. As a result, $a = s(x) = t$, so one has $V_x \subseteq s^{-1}(a)$, showing the openness of $s^{-1}(a)$.

  Then, since all open subsets in A are union of singletons, then any subset $V \subseteq A$ (which is open) automatically has $s^{-1}(V)$ being open (as it's union of $s^{-1}(a)$, $a \in V$). This shows that s is continuous, or $C_A^{\mathsf{Sh}}(U) \subseteq \mathrm{Hom}_{\mathsf{Top}}(U, A)$.

- Suppose $s \in \mathrm{Hom}_{\mathsf{Top}}(U, A)$, then it's clear that all $x \in U$ has $s(x) \in A = F_x$; on the other hand, let $a \coloneqq s(x)$, and consider the open subset $x \in s^{-1}(a) \subseteq U$: Take $a \in F(s^{-1}(a))$, one automatically has all $y \in s^{-1}(a)$ satisfies $s(y) = a = a_y \in F_y$, so the condition for $C_A^{\mathsf{Sh}}(U)$ is trivially satisfied. Hence, $\mathrm{Hom}_{\mathsf{Top}}(U, A) \subseteq C_A^{\mathsf{Sh}}(U)$.

This proves that the constant sheaf $C_A^{\mathsf{Sh}}(\_) = \mathrm{Hom}_{\mathsf{Top}}(\_, A)$ as functor.

2 HD (Exact Sequence Condition in Sheaves)

For this, there are still some more clarifications: For instance, an exact sequence of presheaves, why does it preserve direct limit in general?

Problem 2

(a) For any morphism of sheaves on X, say $\varphi : F \to G$, show that for each point $P \in X$, $\operatorname{Ker}(\varphi)_P \cong \ker(\varphi_P)$, and $\operatorname{Im}(\varphi)_P \cong \operatorname{im}(\varphi_P)$.
(b) Show that φ is injective (resp., surjective) iff the induced map on the stalks φ_P is injective (resp., surjective) for all P.
(c) Show that a sequence $F \to G \to H$ of sheaves and morphisms is exact iff for each $P \in X$ the corresponding sequence of stalks is exact as a sequence of abelian groups.

Solution: Here, we'll use capital letter for the case in sheaves, and lower case letter for the case in abelian groups.

(a) **Kernel:**

For the kernel part it's relatively easy, as ther kernel in $\mathsf{PreSh}(X)$ and $\mathsf{Sh}(X)$ agrees with each other. Consider $\operatorname{Ker}(\varphi)_P = \lim_{U \ni P} \operatorname{Ker}(\varphi(U))$ (where $\operatorname{Ker}(\varphi(U)) \subseteq F(U)$ as set kernel of $\varphi(U)$). Now, consider the following diagram for all $P \in U$:

$$\begin{tikzcd} F(U) \arrow[r, "\varphi(U)"] \arrow[d, "\pi_F"'] & G(U) \arrow[d, "\pi_G"] \\ F_p \arrow[r, "\varphi_P"'] & G_P \end{tikzcd}$$

Which, one has $s_P \in \ker(\varphi_P)$ iff $\varphi_P(s) = 0$, hence there exists some open subset $V \subseteq U$, such that some representative of s_P, say $s \in F(V)$, satisfies $\varphi(s) = 0 \in G(V)$, showing that s_P can be identified as an element of $\operatorname{Ker}(\varphi)_P$ (also, the converse holds, as if it's part of $\operatorname{Ker}(\varphi)_P$, then its representative in some $F(V)$ gets evaluated to 0).

So, set wise one can conclude that $\operatorname{Ker}(\varphi)_P = \ker(\varphi_P)$ (since one can view the direct limit of $\operatorname{Ker}(\varphi)$, as including into F, then take the direct limit).

Image:

For the image part, as the functor is now different, the direct association is harder to draw. Originally, the $\operatorname{im}(\varphi)(U) := \operatorname{im}(\varphi(U))$. However, using similar claims as for the kernel, notice one has $\operatorname{im}(\varphi)_x \cong \operatorname{im}(\varphi_x)$ (since in the category of presheaves, image is a kind of kernel, so the direct limit should be preserved using the same logic).

Now, the sheafification has $\operatorname{Im}(\varphi)(U) := \left\{\text{functions} \quad s : U \to \bigsqcup_{x \in U} \operatorname{im}(\varphi)_x\right\}$ that satisfies other sheafification conditions. In particular, when taking the direct limit $\operatorname{Im}(\varphi)_P$ for any $P \in X$, define the map $\operatorname{ev} : \operatorname{Im}(\varphi)_P \to \operatorname{im}(\varphi)_P \cong \operatorname{im}(\varphi_P)$ by $\operatorname{ev}(s_P) := s(P)$ for its representative $s \in G(U)$ (for some open neighborhood $P \in U \subseteq X$).

First, this is well-defined (as any two functions defined on some open neighborhood of P must agree on some smaller neighborhood of P if they're identified as the same in direct limit, hence the same on P). Second, it's a group homomorphism, as given any $s_P, t_P \in \operatorname{Im}(\varphi)_P$, take some common neighborhood W that their representatives $s, t \in F(W)$ are defined, one has $(s+t)_P = s_P + t_P$, so $\operatorname{ev}(s_P + t_P) = (s+t)(P) = s(P) + t(P) = \operatorname{ev}(s) + \operatorname{ev}(t)$.

Now, to prove it's an isomorphism, let's tackle the two properties respectively:

- If $\operatorname{ev}(s_P) = 0$, then $s(P) = 0$. However, by the definition of sheafification, there exists some open neighborhood $P \in V \subseteq U$ and $t \in G(V)$, such that $s(Q) = t_Q$ for all $Q \in V$; as a result, one has $s(P) = t_P = 0$, so within some open neighborhood $P \in$

$W \subseteq V$, one has $t|_W = 0 \in G(W)$. Hence, one has $s|_W = 0$ also, so $s_P = 0$, showing injectivity of ev function.

- Now, pick any $s'_P \in \mathrm{im}(\varphi)_P$, take its representative $s' \in \mathrm{im}(\varphi(U))$ for some open neighborhood $P \in U \subseteq V$. Now, consider s' as a function $U \to \bigsqcup_{x \in U} \mathrm{im}(\varphi)_P$, one simply has $s'(P) = s'_P$. Which, take s' as a function in $\mathrm{Im}(\varphi)(U)$ and its image in the direct limit, it satisfies what we want (since s' has evaluation of P to be s'_P, this is precisely what we want). Hence, ev is surjective.

With the $\mathrm{ev} : \mathrm{Im}(\varphi)_P \to \mathrm{im}(\varphi)_P \cong \mathrm{im}(\varphi_P)$ being an isomorphism, WLOG one can conclude that $\mathrm{Im}(\varphi)_P = \mathrm{im}(\varphi_P)$ if desired.

(b) One has φ being injective $\iff$ all open subset $U \subseteq X$ has $\varphi(U)$ being injective $\iff$ $\mathrm{Ker}(\varphi)(U) = 0$ for all open subset $U \subseteq X \iff$ its direct limit $\mathrm{Ker}(\varphi)_P = 0$ for all $P \in X$ $\iff \ker(\varphi_P) = 0$ for all $P \in X$, or φ_P is injective (where, the last eqquivalence is given by (a), the other parts are given by the fact that kernels in $\mathsf{Sh}(X)$ and $\mathsf{PreSh}(X)$ coincide).

On the other hand, one has φ being surjective $\iff \mathrm{Im}(\varphi)(U) \cong G(U)$ for all open subset $U \subseteq X \iff$ the direct limit $\mathrm{Im}(\varphi)_P \cong G_P$ for all $P \in X \iff \mathrm{im}(\varphi_P) \cong G_P$ for all $P \in X$, or φ_P is surjective (where, the second equivalence is given by the isomorphism of sheaves and stalks).

(c) Given morphisms of sheaves $f : F \to G$ and $g : G \to H$, one has $F \to G \to H$ being an exact sequence $\iff \mathrm{Im}(f) \cong \mathrm{Ker}(g) \iff$ their direct limit $\mathrm{im}(f_P) \cong \mathrm{Im}(f)_P \cong \mathrm{Ker}(g)_P \cong \ker(g_P)$ for all $P \in X \iff$ the morphisms $f_P : F_P \to G_P$ and $g_P : G_P \to H_P$ has $F_P \to G_P \to H_P$ being exact.

3 ND

Problem 3

(a) Let $\varphi : F \to G$ be a morphism of sheaves on X. Show that φ is surjective iff the following condition holds: for every open set $U \subseteq X$, and for every $s \in G(U)$, there is a covering $\{U_i\}$ of U, and elements $t_i \in F(U_i)$, such that $\varphi(t_i) = s|_{U_i}$ for all i.

(b) Give an example of a surjective morphism of sheaves $\varphi : F \to G$, and an open set U such that $\varphi(U) : F(U) \to G(U)$ is not surjective.

Solution:

(a) $\implies$:

Suppose φ is surjective as a morphism of sheaves, this implies the sheaf $\mathrm{Im}(\varphi) \cong G$ (in particular, given that $\mathrm{Im}(\varphi)(U) = \left\{\text{functions } s : U \to \bigsqcup_{x\in U} \mathrm{im}(\varphi)_x\right\}$)

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Problem 4

(a) Let $\varphi : F \to G$ be a morphism of presheaves such that $\varphi(U) : F(U) \to G(U)$ is injective for each U. Show that the induced map $\varphi^{\mathsf{Sh}} : F^{\mathsf{Sh}} \to G^{\mathsf{Sh}}$ of associated sheaves is injective.

(b) Use part (a) to show that if $\varphi : F \to G$ is a morphism of sheaves, then $\mathrm{Im}(\varphi)$ can be naturally identified with a subsheaf of G.

Solution:

(a) It suffices to show that all the morphisms on the stalks $\varphi_x : F_x \to G_x$ is injective. Yet, this is true based on the fact that direct limit (over the same direct system) preserves exact sequences. Which, in articular if all $\varphi(U)$ is injective, any φ_x is injective.

As a result, the collection of maps for disjoint union $\sqcup \varphi_x : \bigsqcup_{x\in U} F_x \to \bigsqcup_{x\in U} G_x$ is always injective (since injective on each disjoint part).

Now, if considder the morphism $\varphi^{\mathsf{Sh}} : F^{\mathsf{Sh}} \to G^{\mathsf{Sh}}$, if consider $\varphi^{\mathsf{Sh}}(U) : F^{\mathsf{Sh}}(U) \to G^{\mathsf{Sh}}(U)$ by $\varphi^{\mathsf{Sh}}(U)(s) = \sqcup \varphi_x \circ s$ (where $s : U \to \bigsqcup_{x\in U} F_x$ is a function). Then, one has $\varphi^{\mathsf{Sh}}(U)(s) = \varphi^{\mathsf{Sh}}(U)(t)$ iff $s = t$ (since as set map, this is $\sqcup \varphi_x \circ s = \sqcup \varphi_x \circ t$, then with injectivity of $\sqcup \varphi_x$, this enforces $s = t$). So, $\varphi^{\mathsf{Sh}(U)}$ is injective for all open subset $U \subseteq X$, showing φ^{Sh} is injective as a morphism of sheaves.

(b) Given $\varphi : F \to G$ as a morphism of sheaves, then the morphism of presheaves $\mathrm{im}(\varphi) : I \to G$ has $\mathrm{im}(\varphi)(U) : I(U) \hookrightarrow G(U)$ being injective for all open subset $U \subseteq X$ (where $I(U)$ is the set theoretic image of $\varphi(U)$, so I is a presheaf). As a result, part (a) has that its sheafification $\mathrm{im}(\varphi)^{\mathsf{Sh}}(U) : I^{\mathsf{Sh}}(U) \to G^{\mathsf{Sh}}(U) = G(U)$ being injective for every $U \subseteq X$ open.

Here, we denote $\mathrm{Im}(\varphi) := \mathrm{im}(\varphi)^{\mathsf{Sh}}$ (the sheafification of the presheaf of image), hence one has $\mathrm{Im}(\varphi)$ (the image sheaf) being identified as a subsheaf of G.

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Problem 5

Show that a morphism of sheaves is an isomorphism iff it is both injective and surjective.

Solution: $\Longrightarrow$:

First, if $\varphi : F \to G$ is an isomorphism of sheaves, then there exists another morphism of sheaves $\psi : G \to F$, such that $\psi \circ \varphi = \mathrm{id}_F$, and $\varphi \circ \psi = \mathrm{id}_G$. As a result, one has $\psi(U) \circ \varphi(U) = \mathrm{id}_{F(U)}$, and $\varphi(U) \circ \psi(U) = \mathrm{id}_{G(U)}$, so the function $\varphi(U) : F(U) \to G(U)$ is set-theoretic bijective. As a result, one knows φ is injective, and $\mathrm{Im}(\varphi) = G$ (when identified as the set-theoretic subsheaf). So, φ is also surjective.

$\Longleftarrow$:

Now, suppose $\varphi : F \to G$ is both injective and surjective, one has $\mathrm{Ker}(\varphi) = 0$ (which, since it's the kernel of φ as a morphism of presheaf, which indicates that each $\varphi(U) : F(U) \to G(U)$ is injective). On the other hand, one has $\mathrm{Im}(\varphi) \cong G$, which implies that $\mathrm{im}(\varphi)_x \cong \mathrm{Im}(\varphi)_x \cong G_x$